

SEMEY, Kazakhstan

WMO: 361770

Lat: 50.417N

Lon: 80.300E

Elev: 643

StdP: 14.36

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.0	-24.5	-35.7	0.7	-31.0	-29.6	1.1	-24.3	20.9	20.8	18.5	19.7	2.6	90	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	91.6	66.2	88.5	65.2	85.5	64.1	69.0	84.2	67.6	82.4	66.3	80.8	6.4	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.4	92.8	73.9	62.8	87.5	72.4	61.1	82.5	71.8	33.6	84.1	32.5	82.8	31.4	80.8	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	17.8	14.9	13.3	DB	-32.8	98.0	8.0	3.6	-38.5	100.6	-43.2	102.7	-47.7	104.7	-53.5	107.4
(5)				WB	-32.9	72.3	7.8	2.1	-38.5	73.8	-43.1	75.1	-47.5	76.2	-53.2	77.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	40.0	3.6	7.6	23.6	45.4	57.9	68.6	70.8	66.7	55.2	42.3	24.9	11.1	(6)
(7)		DBStd	26.07	15.07	14.42	13.11	9.88	8.27	6.65	5.68	6.50	8.23	8.96	13.46	13.42	(7)
(8)		HDD50	5941	1438	1188	820	198	25	0	0	0	41	269	755	1207	(8)
(9)		HDD65	9638	1903	1608	1284	590	245	43	16	59	309	705	1204	1672	(9)
(10)		CDD50	2277	0	0	0	59	270	557	646	519	196	29	1	0	(10)
(11)		CDD65	500	0	0	0	1	25	150	196	113	15	0	0	0	(11)
(12)		CDH74	6262	0	0	0	39	509	1758	2083	1513	344	16	0	0	(12)
(13)		CDH80	2283	0	0	0	7	146	650	806	566	107	1	0	0	(13)
(14)	Wind	WSAvg	5.6	5.4	5.6	5.9	6.1	6.2	5.7	5.1	4.9	4.7	5.1	5.8	6.3	(14)
(15)	Precipitation	PrecAvg	11.5	0.6	0.6	0.7	0.7	1.1	1.2	1.9	1.0	0.7	1.0	1.0	0.9	(15)
(16)		PrecMax	30.4	1.2	1.7	2.7	2.0	4.6	3.2	5.6	2.9	4.6	3.1	3.3	2.0	(16)
(17)		PrecMin	5.7	0.0	0.1	0.1	0.0	0.0	0.1	0.0	0.1	0.0	0.2	0.0	0.1	(17)
(18)		PrecStd	4.0	0.3	0.4	0.5	0.5	0.9	0.7	1.3	0.7	0.7	0.6	0.6	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.6	36.2	59.3	79.1	90.1	93.9	96.4	95.7	89.1	76.0	57.6	37.7	(19)
MCWB			32.8	32.7	46.4	57.3	62.8	65.8	68.3	67.3	63.4	54.0	44.7	34.0	(20)	
2%		DB	31.3	33.3	50.0	72.7	83.7	90.0	91.3	90.3	82.7	68.3	50.3	34.8	(21)	
		MCWB	28.4	29.9	42.0	53.9	60.4	65.0	67.5	65.8	60.2	51.6	42.2	31.6	(22)	
5%		DB	27.7	30.8	43.8	68.2	79.8	87.0	87.8	86.2	77.5	62.9	45.7	31.7	(23)	
		MCWB	24.9	27.8	38.3	52.0	58.6	64.2	66.2	64.5	58.1	49.3	39.3	28.8	(24)	
10%		DB	24.0	27.7	39.3	63.5	75.5	83.7	84.6	82.3	72.5	57.1	40.8	28.8	(25)	
		MCWB	21.7	25.0	35.2	49.3	57.1	63.3	65.1	62.9	55.7	46.3	36.2	26.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.0	33.6	47.9	58.8	65.0	69.9	72.3	70.0	64.4	55.3	47.4	34.8	(27)
MADB			35.2	35.7	57.8	74.4	84.3	86.0	88.8	87.4	85.6	71.7	53.1	37.1	(28)	
2%		WB	28.6	30.4	42.4	55.6	62.3	67.8	69.7	67.7	61.1	52.3	42.3	31.9	(29)	
		MADB	31.1	32.9	48.9	69.9	79.4	82.7	84.1	84.4	78.9	65.8	48.9	34.5	(30)	
5%		WB	25.0	27.8	38.5	52.8	60.3	66.3	68.3	66.1	58.9	49.8	39.3	28.9	(31)	
		MADB	27.5	30.6	43.4	66.0	75.5	81.5	82.5	82.2	74.6	60.1	45.2	31.6	(32)	
10%		WB	21.7	24.9	35.5	50.2	58.4	64.9	67.1	64.5	56.6	47.2	36.3	26.3	(33)	
		MADB	24.0	27.8	39.0	62.0	73.0	79.7	80.6	79.2	69.8	55.7	40.7	28.8	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.3	18.9	17.4	22.7	24.8	23.2	22.1	25.2	26.2	20.3	14.7	15.1	(35)
MCDBR			14.5	15.7	20.4	31.8	32.8	28.8	28.4	32.5	34.8	32.0	20.2	15.1	(36)	
MCWBR			13.4	14.0	13.6	16.1	13.4	9.8	9.4	12.3	15.2	16.9	13.1	13.5	(37)	
5% WB		MCDWR	14.7	15.4	19.5	29.3	27.8	24.9	23.8	28.5	31.5	27.9	18.5	15.2	(38)	
		MCWBR	13.8	14.0	13.5	15.5	12.8	9.5	8.9	11.8	14.6	15.7	12.7	13.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.304	0.344	0.399	0.424	0.424	0.422	0.444	0.414	0.370	0.361	0.346	0.313	(40)	
taud		2.037	1.955	1.950	2.099	2.148	2.240	2.187	2.246	2.314	2.235	2.112	2.008	(41)		
Ebn at Noon		216	238	247	258	264	266	257	259	259	234	199	188	(42)		
Edh at Noon		27	39	47	46	46	42	44	39	33	30	26	24	(43)		
(44)	All-Sky Solar Radiation	RadAvg	462	813	1230	1533	1975	2095	1914	1724	1290	721	383	326	(44)	
(45)		RadStd	32	54	107	116	132	144	131	82	101	87	39	35	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page